

Lecture 2

Exercise, Linearization of system with multiple states

Consider a system

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1 + x_1^3 + x_2^2\end{aligned}$$

Questions:

1. What is the set of all equilibria of the system
2. What is the linearization of the system at $(0,0)$ ← the origin

Answer:

1. The system

$$\dot{x}_1 = x_2 = f_1(x, u)$$

$$\dot{x}_2 = -x_1 + x_1^3 + x_2^2 = f_2(x, u)$$

In this case:

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad u = \emptyset$$

Equilibria (x_{10}, x_{20}) are points where

$$\dot{x}_1 = f_1(x_0, u_0) = f_1(x) = 0$$

$$\dot{x}_2 = f_2(x_0, u_0) = f_2(x) = 0$$

Hence:

$$\dot{x}_1 = x_2 = 0 \Rightarrow x_{20} = 0$$

$$\dot{x}_2 = -x_1 + x_1^3 + x_2^2 = -x_1 + x_1^3 + x_{20}^2 = 0$$

$$= -x_1 + x_1^3 + 0 = 0$$

$$= -x_1(1 - x_1^2) = 0$$

$$= -x_1(1 - x_1)(1 + x_1) = 0$$

$$x_1 = 0 \vee x_1 = 1 \vee x_1 = -1$$

Therefore the equilibria are:

$$(x_0, x_{20}) = (0, 0), (1, 0), (-1, 0)$$

There are 3 equilibria of the system.

2. Linearization at (0,0)

$$\dot{x} = f(x, u) \Rightarrow \dot{x} = Ax + Bu$$

use jacobian

$$A = \left. \frac{\partial f}{\partial x} \right|_{x_0, u_0}, \quad B = \left. \frac{\partial f}{\partial u} \right|_{x_0, u_0}$$

Because the system has no input, then $B=0$

Because the system has 2 states, x_1 and x_2 ,
it means $x \in \mathbb{R}^2$ then $A \in \mathbb{R}^{2 \times 2}$ (a matrix)

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} \left(x_{10}, x_{20} \right) \quad \leftarrow \text{jacobian of } \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \text{ with respect to } \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\left. \frac{\partial f_1}{\partial x_1} \right|_{x_0} = \left. \frac{\partial (x_2)}{\partial x_1} \right|_{(x_{10}, x_{20})} = 0$$

$$\left. \frac{\partial f_1}{\partial x_2} \right|_{x_0} = \left. \frac{\partial (x_2)}{\partial x_2} \right|_{(x_{10}, x_{20})} = 1$$

$$\left. \frac{\partial f_2}{\partial x_1} \right|_{x_0} = \left. \frac{\partial (-x_1 + x_1^3 + x_2^2)}{\partial x_1} \right|_{(x_{10}, x_{20})} = -1$$

$$\left. \frac{\partial f_2}{\partial x_2} \right|_{x_0} = \left. \frac{\partial (-x_1 + x_1^3 + x_2^2)}{\partial x_2} \right|_{(x_{10}, x_{20})} = 2x_2 \big|_{(x_{10}, x_{20})}$$

Therefore

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 2x_2 \end{bmatrix} \left(x_{10}, x_{20} \right)$$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -x_1 \end{cases}$$

at $(x_{10}, x_{20}) = (0, 0)$

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Hence, the linearized model

$$\dot{x} = Ax = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -x_1 \end{bmatrix}$$